

KLYUCHI KAMCHATKA, Russian Federation

WMO: 323890

Lat: 56.317N

Lon: 160.833E

Elev: 29

StdP: 100.98

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.8	(c) -30.4	(d) -35.2	(e) 0.1	(f) -32.7	(g) -32.9	(h) 0.2	(i) -30.4	(j) 10.5	(k) -10.1	(l) 9.4	(m) -9.2	(n) 0.9	(o) 110	(p) 0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 25.3	(d) 18.1	(e) 23.5	(f) 17.2	(g) 21.7	(h) 16.3	(i) 19.1	(j) 23.7	(k) 17.9	(l) 22.4	(m) 16.8	(n) 20.8	(o) 2.3	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.3	21.6	15.9	11.4	20.5	14.9	10.6	19.1	54.5	23.7	50.8	22.4	47.3	20.9	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.4	(b) 9.0	(c) 7.8	(d) DB	(e) -35.3	(f) 27.7	(g) 3.3	(h) 1.1	(i) -37.6	(j) 28.5	(k) -39.5	(l) 29.2	(m) -41.4	(n) 29.8	(o) -43.8	(p) 30.6	(q) 25.0
(a) 10.4	(b) 9.0	(c) 7.8	(d) WB	(e) -35.3	(f) 20.9	(g) 3.3	(h) 1.6	(i) -37.7	(j) 22.0	(k) -39.6	(l) 22.9	(m) -41.5	(n) 23.8	(o) -43.9	(p) 25.0	(q) 25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-15.6	-13.4	-7.6	-1.4	5.8	12.5	16.2	15.1	9.8	2.7	-5.6	-12.9
	DBStd	12.28	9.49	7.37	5.81	3.75	3.41	3.61	3.05	2.57	2.99	3.68	5.35	7.84
	HDD10.0	3936	794	657	547	343	138	15	1	0	38	228	467	709
	HDD18.3	6515	1052	890	806	593	388	176	79	107	254	486	717	967
	CDD10.0	484	0	0	0	0	8	91	193	157	34	0	0	0
	CDD18.3	22	0	0	0	0	0	3	14	5	0	0	0	0
	CDH23.3	161	0	0	0	0	1	30	105	25	0	0	0	0
	CDH26.7	8	0	0	0	0	0	2	6	1	0	0	0	0
Wind	WSAvg	3.0	2.7	3.3	3.5	3.4	2.9	2.7	2.3	2.4	2.9	3.3	3.8	3.2
Precipitation	PrecAvg	618	68	58	42	30	37	40	49	68	52	54	60	60
	PrecMax	881	145	120	111	87	87	119	98	185	176	146	121	122
	PrecMin	456	9	14	0	3	8	2	6	17	2	0	7	21
	PrecStd	114	33	32	25	18	18	28	24	33	29	33	32	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	1.5	2.7	9.2	21.1	26.1	27.4	26.1	19.7	11.6	5.3	3.0
		MCWB	0.6	0.4	0.4	4.5	12.7	16.4	19.6	20.2	14.7	7.0	3.2	2.1
	2%	DB	0.6	0.4	1.8	6.5	17.5	24.0	25.9	23.5	17.7	10.1	3.7	0.6
		MCWB	-0.1	-0.4	0.0	2.4	10.4	15.8	18.8	18.0	13.3	6.2	2.0	-0.1
	5%	DB	-0.5	-0.6	0.9	4.6	15.0	21.8	24.4	21.6	16.1	8.8	2.4	-0.3
		MCWB	-1.0	-1.3	-0.5	1.1	8.9	14.7	18.0	16.8	12.2	5.8	0.9	-0.7
	10%	DB	-2.2	-2.5	0.1	3.3	12.5	19.6	22.7	20.0	14.6	7.6	1.2	-1.8
		MCWB	-2.7	-3.3	-1.1	0.4	7.4	13.5	17.2	15.7	11.3	5.2	-0.1	-2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	0.7	1.2	4.8	12.8	18.3	21.0	20.8	15.5	8.5	3.8	2.2
		MCDB	1.2	1.3	1.8	8.6	20.5	23.8	26.0	25.1	18.8	10.1	4.8	3.0
	2%	WB	0.1	-0.2	0.4	2.7	10.6	16.4	19.5	18.8	13.9	7.2	2.4	0.2
		MCDB	0.4	0.3	1.2	5.8	17.1	22.3	24.4	22.6	16.4	8.9	3.3	0.4
	5%	WB	-1.0	-1.2	-0.2	1.5	9.1	15.2	18.5	17.3	12.7	6.4	1.3	-0.9
		MCDB	-0.5	-0.6	0.8	3.8	14.6	20.9	23.3	20.8	15.1	8.2	2.0	-0.4
	10%	WB	-2.7	-3.3	-1.1	0.8	7.6	13.9	17.5	16.2	11.8	5.4	0.1	-2.5
		MCDB	-2.1	-2.5	0.1	2.7	12.2	19.1	22.1	19.2	13.9	7.3	1.0	-1.8
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.4	8.1	7.0	8.4	9.1	8.8	7.8	7.4	5.7	5.5	6.2
		MCDBR	5.3	5.1	5.7	8.5	13.4	13.8	12.8	10.9	9.6	7.2	5.3	6.0
		MCWBR	5.0	4.8	4.2	5.6	7.8	7.1	6.7	6.1	5.6	4.7	4.3	5.7
	5% WB	MCDBR	4.9	5.1	4.7	7.1	12.9	12.3	11.5	10.0	8.2	6.0	4.9	5.6
		MCWBR	5.0	5.0	3.8	4.9	7.7	6.9	6.5	6.0	5.1	4.3	4.2	5.5
Clear-Sky Solar Irradiance	taub	0.273	0.310	0.368	0.433	0.490	0.460	0.479	0.454	0.372	0.342	0.286	0.265	
	taud	2.047	2.019	1.881	1.771	1.786	2.044	2.053	2.118	2.330	2.209	2.103	2.006	
	Ebn at Noon	591	717	757	764	747	784	758	748	773	682	569	486	
	Edh at Noon	62	97	145	189	199	156	152	134	94	81	58	50	
All-Sky Solar Radiation	RadAvg	0.66	1.50	2.90	4.53	5.32	5.21	5.10	3.88	2.91	1.62	0.79	0.45	
	RadStd	0.07	0.18	0.29	0.37	0.49	0.46	0.46	0.38	0.28	0.14	0.05	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page